



Indian Institute of Technology Jodhpur
Office of Academics
 NH65, Nagaur Road, Karwar, Jodhpur 342037, INDIA

The course and timings-

Course Number and Title	MEL1010 Engineering Mechanics (Jan-May 2026)	
Lectures	Days	Monday, Wednesday
	Time	Non-English (Conversational Language) (Monday, 5:00 PM- 5:50 PM) (Wednesday, 1 PM-1:50 PM) (G Slot) English (Monday, 8:00 PM- 8:50 AM) (Wednesday, 8 AM- 8:50 AM) (K Slot)
	Venue	Non-English (Conversational Language): LHB205 (G Slot), English: LHB110 (K Slot)
Tutorials	Days	1
	Time	Non-English (Conversational Language) (Friday, 1 PM - 1:50 PM) (G Slot) English (Friday, 8 AM – 8:50 AM) (K Slot)
	Venue	To be decided (different for different groups)
Laboratory	Days	NA
	Time	NA
	Venue	NA

Instructor(s)-

Name	Email	Department	Contact Number	Office Address
Course Coordinator				
Dr. B. Ravindra	ravib@iitj.ac.in	Mechanical	0291-280-1503	Office No. 221, Department of Mechanical Engineering, IIT Jodhpur
Co-Instructor				
Dr. Aali Pant	aalipant@iitj.ac.in	Civil and Infrastructure Engineering	0291-280-1664	Office No. 121, Department of Civil and Infrastructure Engineering, IIT Jodhpur
Tutors				
Prof. Nitesh Arora	nitesharora@iitj.ac.in	Mechanical	0291 280 1536	Office No. 204, Manufacturing building, ME, IIT Jodhpur
Dr. Jayant Kumar Mohanta	jayant@iitj.ac.in	Mechanical	0291 280 1524	Office No. 322, Department of Mechanical Engineering, IIT Jodhpur
Dr. Ambuj Kumar Gautam	ambujk@iitj.ac.in	Mechanical	0291-280-1540	Office No. 202, Department of Mechanical Engineering, ME&CT building, IIT Jodhpur

Dr. Babu Ranjan Thakur	brthakur@iitj.ac.in	Mechanical	0291 280	Office No. , Department of Mechanical Engineering, IIT Jodhpur
Dr. Barun Pratiher	barun@iitj.ac.in	Mechanical	0291 280 1505	Office No. 223, Department of Mechanical Engineering, IIT Jodhpur
Dr. Bhupendra Singh	bhupendra@iitj.ac.in	Civil and Infrastructur e Engineering	0291-280-1659	Office No. 119, Department of Civil and Infrastructure Engineering, IIT Jodhpur

Office Timings during the week: 9AM to 6 PM Call faculty members and make an appointment

Contact details of Teaching Assistants (TAs): Please contact the TAs for any query related to the course or attendance.

Name of the TA	Roll No	Contact details
Akash Vyas	D21ME101	9717191841
Manoj Paul	D22ME101	6268375525
Jay Hitendrakumar K.	P21ME202	7405138075
Akhilesh Gupta	P25ME0002	
Shamir Khan	P23ME0009	9887035534
Kshitij Saren	M24MEA0005	
Sakshi Khiroriya	M25MER015	7880241687
Manisha K.	M25MER013	8858903414
Vaibhav N. S.	M25MEA010	7459822781
Riya Pankaj	M25MEA007	
Dhirendra	M25MEA004	6367482787
Ashish Gupta	P21CI003	
Kanika	P23CI0012	8219594964
Yashwant Choudhary	P23CI0011	9413846006
Anish Sharma	P23CI0007	8219596735
Mohsin Jamil Md Nasir	P23CI0004	
Ankit Kumar Maurya	P22CI001	8127732923
Lalit Bahal	P24CI0003	

Lecture-wise detailed course content as approved by the Senate:

Title	Engineering Mechanics	Number	MEL1010
Department	Mechanical Engineering	L-T-P [C]	2-1-0 [3]
Offered for	B. Tech. 1st Year (All Branches)	Type	Compulsory
Prerequisite	Nil		

Objectives

1. To apply knowledge of statics and dynamics to particles and rigid bodies

Learning Outcomes

1. To model practical structural problems using concepts of free body diagrams and equilibrium conditions
2. To solve dynamic problems using Newton's Law, Work-Energy and Impulse-Momentum Principles

Course Content

Statics [10 Lectures]: Basic dimensions in Mechanics, Law of dimensional homogeneity, Vector and Scalar Quantities, Elements of vector algebra. Moment of a force about a point/axis, Couple, Free Body Diagram, Equations of Equilibrium, Static indeterminacy, Trusses and Beams, Coulomb Friction, Surface contact friction, Transmission of power through belt. Pappus-Guldinus Theorems, second moments and product of Area, Principal axes.

Dynamics of Particles [6 Lectures]: Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion. Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles

Dynamics of Rigid bodies [12 Lectures]: Translation/Rotation of rigid bodies, Chasles' theorem, time derivative of vector for different references. Parallel axis theorems, Rotational Pure rotation of a body of revolution about its axis of revolution/combined with translation. Three-dimensional rotation, moment of inertia tensor, relation between angular momentum and torque in three dimensions, Motion in non-inertial frames, centrifugal force, Coriolis force/acceleration, rate of change of vector in inertial and rotating frames.

Textbooks:

1. Meriam, J. L., Kraige, L. G., (2002), Engineering Mechanics, John Wiley & Sons

Reference Books:

1. Shames, I., (2003), Engineering Mechanics, Prentice Hall
2. Gross, D., Hauger, W. and Schröder J., (2012), Engineering Mechanics, Springer
3. Harbola, M K., (2013), Engineering Mechanics, CENGAGE Learning

Online Course Material:

1. Harbola, M K., Engineering Mechanics, NPTEL Course Material, Department of Physics, Indian Institute of Technology Kanpur, <https://nptel.ac.in/courses/115104094/>
2. Lewin, W. H. G, Physics I: Classical Mechanics - Fall 1999, MIT, http://videlectures.net/mit801f99_physics_classical_mechanics/

Evaluation policy for the course:

Components	Weightage (%)	Date and Timings	Remarks
Assignments	15%	Every Friday (Tutorial Class) Tutorial Assignment will be posted in Google Classroom by Friday night of previous week	
Quizzes	15 %	Dates to be announced	
Minor	20%	Syllabus: Statics	Written Examination
Major	40%	Syllabus: Complete	Closed Book, Written examination, Scientific calculators are permitted
Project	10%	To be announced later	Prepare Spreadsheets
	100%		

Guidelines for Examination:

1. Scientific calculators are allowed with no internet. Smart watches and mobiles are not allowed.
2. If anyone is found doing malpractice in examination, answer sheets will be taken. Further action will be taken as per Institute rules.

Attendance Policy:

1. Students are expected to attend all the lectures and tutorial classes
2. Institute attendance policy will be followed.

Any other relevant information:

1. Bring Scientific calculator in tutorial session, lecture classes, minor examination and major examination.
2. Solve tutorial assignments very sincerely. It will help you to clarify your concepts of Engineering Mechanics.
3. In the beginning, you may find the course very easy as you will be studying some concepts which you have already studied in the 11th or 12th classes. But study regularly, the course will require you to solve many numerical problems.
4. You should refer to class notes or study from textbooks. Apart from solving tutorial questions, you should solve as many numerical problems (solved and unsolved from textbooks or reference books) as you can.